

Project SEN124A (2025–2026) Due by 8 April 2026, 23:59

Designing a Regional Public Service Hub in the Randstad

Project Description

Large metropolitan regions increasingly rely on centralized public service hubs such as government service centers, emergency operation centers, (non-hospital) healthcare coordination facilities to coordinate essential services and emergency response. In polycentric regions like the Randstad, deciding where to locate such a hub is far from trivial. While geographic centrality may appear intuitive, real accessibility is shaped by the structure of the underlying transportation network, population distribution, and the existence of alternative routes.

The Dutch government is planning to establish a new regional public service hub in the Randstad. Due to financial and organizational constraints, only a single location can be selected. The hub must serve residents across the region and remain accessible under a variety of conditions. Decision-makers are aware that a poorly chosen location could lead to long travel times for large parts of the population or create structural dependencies on a limited number of network connections. Your group will advise the government as a network analyst. Using network theory, you will assess the Randstad transportation network, evaluate alternative locations for the hub, and assess if your recommendation remains valid when assumptions or network conditions change. The project is structured in four parts, each building on the previous one.

To complete the following components of this project, the dataset including the Python notebook and Gephi file will be available on Brightspace. You may source additional data to complete the tasks and develop a more comprehensive and detailed report.

Part 1: Understanding the Baseline Network

Before selecting any location, decision-makers want to understand the structural properties of the Randstad network itself. You can assume that the Randstad consists at least of the following cities/locations: Zaanstad, Amsterdam, Haarlem, Schiphol, Leiden, Den Haag, Zoetemeer, Delft, Rotterdam, Spijkenisse, Dordrecht, Vlaardingen, Alphen aan de Rijn, Gouda, Woerden, Utrecht, Hilversum, Almere, and Amersfoort. You may add more cities if you like. **Use the same network for all the following parts.**

They are not interested in specific hub locations yet. First, they would like to identify which cities or nodes are structurally well-positioned from a network perspective under business-as-usual conditions.

In this part, you will work with a baseline network of the Randstad that includes cities as nodes and major transportation links as edges, weighted by distance or travel time. Your task is to explore this network using **at least three network metrics**. The selected

metrics should capture different aspects of network performance, such as accessibility, structural importance, or vulnerability.

You must calculate these metrics and **motivate why they are appropriate** for evaluating a public service hub that is intended to serve a large and diverse population.

In this part, your main tasks are as follows:

- Describe the baseline network and its assumptions
- Select and calculate at least three network metrics
- Motivate why these metrics are suitable in this societal context
- Interpret the results using network theory and visualizations

Part 2: Identifying and Comparing Candidate Hub Locations

Building on the baseline analysis, decision-makers now ask you to move from general network understanding to concrete location choices. Rather than immediately selecting a single location, they want to explore a small set of alternative locations. Based on the insights from Part 1, you will identify **at least three candidate locations** that could realistically host the public service hub.

For each candidate location, you will evaluate how well the hub performs in terms of accessibility for residents across the Randstad. For this task you should consider translating structural metrics that you used in Part 1 into performance measures. For example, you can weigh the metrics with population values to represent population-weighted travel times or accessibility distributions. The aim is not to find a universally “best” solution, but to understand how different locations perform differently, and which groups benefit or are disadvantaged by each option.

This part focuses on comparison and trade-offs. A location that performs well on average may still leave areas underserved, while another location might offer more balanced accessibility at the cost of longer overall travel times.

In this part, your main tasks are:

- Identify at least three alternative hub locations
- Evaluate their performance using the network metrics from Part 1
- Use at least two different transport modalities to identify alternative locations
- Compare outcomes and explicitly discuss trade-offs
- Critically reflect on how the selected metrics support the comparison

- Use visualizations and quantitative evidence to support your analysis

Part 3: Testing Robustness of Alternative Scenarios

Real-world decisions are rarely made under stable conditions. Infrastructure networks are subject to disruptions, growth, and changing usage patterns. Decision-makers therefore want to know if your proposed hub location remains a good choice when conditions change.

In this part, you will define one realistic alternative scenario and re-evaluate the candidate hub locations under this scenario. Examples include temporary road closures, increased congestion on specific corridors, or population growth in certain subregions. The scenario must be clearly defined and justified, and its effects should be analyzed using the **same network metrics** that you used in the previous parts.

The goal is to assess if the relative performance of the candidate locations changes, and if your recommendation remains robust or becomes sensitive to specific assumptions.

In this part, your main tasks are:

- Define and justify one alternative scenario
- Based on the alternative scenario modify the network or its weights accordingly
- Reassess the alternative locations using the same network metrics
- Discuss how and why the ranking or performance changes

Part 4 [Bonus]: Designing an Improved Network

Decision-makers are open to long-term investments that could improve the performance of the public service hub. As a bonus task, you can explore if **adding or strengthening a connection** between two cities could significantly improve network efficiency and accessibility. In this part, you will design a small modification to the network for example, you can suggest adding a new connection or reducing travel time on an existing one and evaluate how this affects the performance of your preferred hub location. This exercise shifts the focus from selecting the best location within a given network to co-designing the network and the hub location together. In other words, the network modifications you propose in this part is solely new to the network and differs from the scenarios you proposed from part 3 (e.g. differs from road closure).

In this part (optional), you may:

- Propose one new or improved network connection

- Justify the design choice using network theory
- Evaluate its impact on accessibility and hub performance

Structure of the report:

You will submit a report of **no more than 15 pages**, excluding appendices. Your report should be structured as follows:

- 1. Introduction and Problem Statement:** Explain the importance of accessibility and the objectives of your analysis (max 500 words)
- 2. Part 1: Understanding the Baseline Network**
- 3. Part 2: Identifying and Comparing Candidate Hub Locations**
- 4. Part 3: Testing Robustness of Alternative Scenarios**
- 5. Part 4 [Bonus]: Designing an Improved Network**
- 6. Discussion and Recommendations:** Summarize your key findings, discuss limitations, and offer strategic recommendations.
- 7. References**
- 8. Appendix**
- 9. Hand in the code**

Assessment and Grading Philosophy

This project will be assessed holistically. The grading does **not** follow a traditional checklist or item-by-item rubric. The evaluation focuses on the quality, coherence, and depth of your network analysis, as well as your ability to translate network-theoretic insights into meaningful conclusions.

Baseline Expectations

A project that:

- completes Parts 1–3,
- correctly applies network metrics,
- provides clear explanations and visualizations,
- and presents a coherent narrative from analysis to recommendation,

will receive a solid passing grade.

Completing **Part 4 (bonus)** in a meaningful way raises your grade but does not automatically guarantee a high mark. Depth and justification matter more than simply including an extra section.

Higher Grades: The following aspects strongly influence the final grade and may help you to get higher grade:

1. Quality of Network Reasoning: Grades increase when network metrics are:

- well-chosen for the societal problem,
- correctly interpreted (not just reported),
- including critical reflections and limitation.

2. Depth of Comparative Analysis: For Parts 2 and 3, strong projects go beyond stating which location performs best. They:

- clearly explain why differences emerge from the network structure,
- identify trade-offs between alternative locations,
- use elaborate visualizations,
- and show how these trade-offs depend on the chosen metrics and assumptions.

Projects that explicitly reassess earlier conclusions considering new scenarios are graded more highly.

3. Thoughtful Use of Scenarios and Assumptions: Your creativity will be rewarded when it is purposeful. For example,

- exploring alternative hub locations that are not obvious but well-justified,
- designing realistic and analytically relevant scenarios that are justified with literature review,
- finding your own data or refining the network in a way that deepens insight.

4. Bonus Part – Designing the Network: Part 4 is an opportunity to demonstrate advanced understanding. High-quality bonus work:

- proposes network modifications grounded in network theory,
- clearly explains how and why the intervention improves performance,
- analytical and quantitative evidence is used for the interventions,

- and links the intervention back to the part 2-3 (i.e., chosen hub location(s)).

A superficial network modification (e.g. adding a random connection “just to do Part 4”) will not get you higher the grade.

5. Coherence, Clarity, and Polish: Projects are also evaluated on:

- clarity of writing and structure,
- transparency of assumptions,
- consistency between parts,
- and clear high-quality presentation of figures and arguments.

A technically strong analysis can still receive a lower grade if the reasoning is not communicated well.

Final Grade Interpretation

In general:

- Projects that complete Parts 1–3 competently and coherently will score in the **passing to mid-range**.
- Projects that show **strong analytical depth, critical reflection, and creative yet disciplined use of network theory** will score **higher**.
- In addition to all the previous points, projects that meaningfully integrate Part 4 and revisit earlier conclusions considering new insights are likely to receive the **highest grades**.

The final grade reflects the **overall contribution of the project as a piece of network-based analysis**, not the number of elements included.